

(0000CJA102120027)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

AIOT

18-04-2021

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE**ANSWER KEY****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	A	A	C	C	D	A	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	B	A	C	C	C	B	D	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2.00	30.00	45.00	37.50	25.00	5.00	21.70 to 21.75	2.20 to 2.25	6.41	4.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	C	B	D	A	D	A	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	B	A	B	B	D	B	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3.00	4.00	140.00	281.00	3.00	2.50	5.00	4.00	125.00	6.00

PART-3 : MATHEMATICS

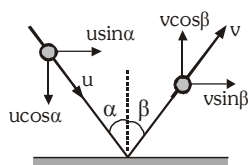
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	D	B	B	D	A	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	D	C	B	A	B	C	D	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	8.00	4.00	4.00	4.00	0.33 or 0.34	7.00	13.33 or 13.34	2.50	58.00	1.00

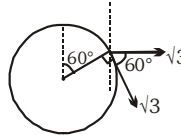
JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (D)**2. **Ans. (C)**3. **Ans. (A)**

Sol. $W = 1 \times 10 \times 8 + 0.1 \times 10 \times 2$
 $+ 0.2 \times 10 \times 2 + 0.4 \times 10 \times 1$
 $= 80 + 2 + 4 + 4$
 $= 90 \text{ J}$

4. **Ans. (A)**

Sol. $v \sin \beta = u \sin \alpha$ and $e = \frac{v \cos \beta}{u \cos \alpha} \Rightarrow e = \frac{\tan \alpha}{\tan \beta}$

5. **Ans. (C)**

Sol. 
 $\sqrt{(\sqrt{3})^2 + (\sqrt{3})^2 + 2(\sqrt{3})(\sqrt{3})\cos 60^\circ} = 3 \text{ m/s}$

6. **Ans. (C)**

Sol. $\frac{y}{x} = \tan 37^\circ = \frac{3}{4}$

$$\Rightarrow y = \frac{3x}{4}$$

$$F = T \cdot 2(2y)$$

$$= 4T \times \frac{3x}{4}$$

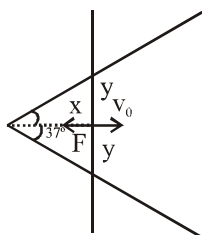
$$F = 3Tx \text{ (Along -ve x)}$$

from work energy theorem

$$\int_{x_0}^{2x_0} (-3Tx) dx = 0 - \frac{1}{2}mv_0^2$$

$$\Rightarrow -3T \cdot \left(\frac{x^2}{2} \right)_{x_0}^{2x_0} = -\frac{mv_0^2}{2}$$

$$\Rightarrow 9x_0^2 T = mv_0^2 \Rightarrow v_0 = 3x_0 \sqrt{\frac{T}{m}}$$

7. **Ans. (D)**

Sol. AB. $w_1 = \text{Area} = \frac{3}{2}P_0V_0$

$$\Delta U_1 = \frac{3}{2}nR(T_B - T_A) = \frac{3}{2}(P_B V_B - P_A V_A)$$

$$= \frac{9}{2}P_0V_0$$

$$\therefore Q_1 = \Delta U_1 + W_1 = 6P_0V_0$$

BC $W_2 = 0$ and $Q_2 = -ve$

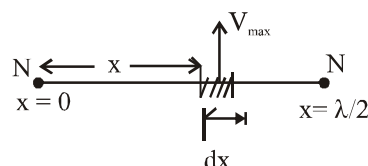
CA $W_3 = -P_0V_0$ (Area) and $Q_3 = -ve$

$$\therefore \eta = \frac{w_1 + w_2 + w_3}{Q(+ve)} = \frac{\frac{3}{2}P_0V_0 + 0 - P_0V_0}{6P_0V_0}$$

$$= \frac{1}{12}$$

8. **Ans. (A)**

Sol. $y = A \sin kx \sin \omega t$

For an element at position x ,

$$v = \frac{\partial y}{\partial t} = A \sin kx \cos \omega t \cdot \omega$$

$$v = (\omega A) \sin kx \cos \omega t$$

$$\therefore v_{\max} = (\omega A) \sin kx = v_0 \sin kx$$

At antinode $|\sin kx| = 1 \therefore v_0 = \omega A$

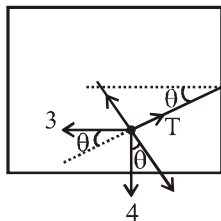
Now $dk_{\max} = \frac{1}{2}dm \cdot v_{\max}^2$

$$= \frac{1}{2} \cdot \mu dx \cdot v_0^2 \sin^2 kx$$

$$\therefore k_{\max} = \frac{\mu v_0^2}{2} \int_0^{\lambda/2} \sin^2 kx dx = \frac{\mu v_0^2 \lambda}{8}$$

9. Ans. (B)

Sol.



$$3 \sin \theta = 4 \cos \theta$$

$$\Rightarrow \tan \theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$$

10. Ans. (C)

Sol. From Gauss law

$$E \cdot 4\pi r^2 = 4\pi G \int \rho dv$$

$$\Rightarrow E = \frac{G}{r^2} \int_0^r \rho_0 \left(1 + \frac{r}{R}\right) \cdot 4\pi r^2 dr$$

$$= \frac{G}{r^2} \cdot 4\pi \rho_0 \int_0^r \left(r^2 + \frac{r^3}{R}\right) dr$$

$$= \frac{4\pi \rho_0 G}{r^2} \cdot \left(\frac{r^3}{3} + \frac{r^4}{4R}\right) = 4\pi \rho G \left(\frac{r}{3} + \frac{r^2}{4R}\right)$$

$$\therefore \text{at } r = \frac{R}{2}$$

$$E = 4\pi \rho_0 G \left(\frac{R}{6} + \frac{R}{16}\right)$$

$$E = \left(\frac{11\pi}{12}\right) \rho_0 GR$$

11. Ans. (C)

Sol. Let θ = the temperature of the body,
 θ_0 = the temperature of the surroundings
 = 20°C (given).

$$\text{Rate of loss of heat} = \frac{dQ}{dt} = k(\theta - \theta_0)$$

$$\text{For } \theta = 40^\circ\text{C}, \frac{dQ}{dt} = 10 = k(40 - 20) \Rightarrow k = \frac{1}{2}$$

When the temperature of the body is raised uniformly, its temperature is given

$$\text{by } \theta = 20 + \left(\frac{80 - 20}{300s}\right) t = 20 + \frac{t}{5},$$

where t = time in seconds

$$\therefore \frac{dQ}{dt} = k(\theta - \theta_0) = \frac{1}{2} \left(\frac{t}{5}\right) = \frac{t}{10}$$

$$\therefore Q = \int_0^{300} \frac{t}{10} dt = \frac{1}{20} (300)^2 J = 4500J$$

12. Ans. (A)

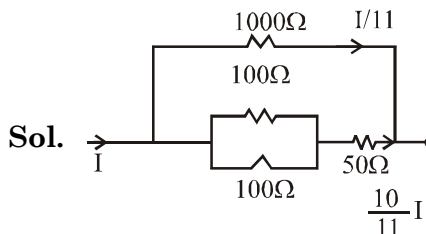
Sol. Both cuts equal number of electric lines

13. Ans. (B)

$$\text{Sol. } \frac{kq}{R} + \frac{kQ}{2R} + \frac{k(2Q)}{3R} = 0$$

$$q = -\frac{7Q}{6}$$

14. Ans. (A)



Sol.

$$I_g = \frac{5}{11} I$$

$$I = \frac{11}{5} I_g$$

$$I_{\max} = \frac{11}{5} \cdot 10 = 22 \text{ mA}$$

15. Ans. (C)

Sol. Equivalent resistance = 6Ω

if current is I , current through $8\Omega = \frac{I}{3}$.

$$\text{now, } P_{\text{total}} = 6I^2 \quad \dots (1)$$

$$P = \left(\frac{I}{3}\right)^2 \cdot 8 = \frac{8}{9} I^2 = \text{Power through } 8\Omega.$$

$$\text{Now, heat develops} = \frac{P}{P_{\text{total}}} \left(\frac{Q^2}{2c}\right) = \frac{2Q^2}{27c}$$

16. Ans. (C)

$$\text{Sol. } I \left(\omega \frac{d\omega}{d\theta}\right) = MB \sin \theta$$

$$M = NIA$$

17. Ans. (C)

Sol. Since flux decreasing field line will be such that it will increase the 'B'.

18. Ans. (B)

$$\text{Sol. } \frac{1}{v} - \frac{1}{2h} = -\frac{2}{h}$$

$$v = \frac{-2h}{3}$$

distance of its image from observer

$$= \frac{d - \frac{2h}{3}}{\mu}$$

apparent distance between these two

$$= \frac{d - \frac{2h}{\mu}}{\mu} - \frac{d - 2h}{\mu} = \frac{4h}{3\mu}$$

19. **Ans. (D)**

Sol. $A_1 \xrightarrow{\lambda_1} A_2 \xrightarrow{\lambda_2}$

we have rate equation

$$\frac{dN_{A_2}}{dt} = \lambda_1 N_{A_1} - \lambda_2 N_{A_2}$$

solving the equation gives N_{A_2} which is

$$\text{maximum at } t = \frac{\ln(\lambda_2 / \lambda_1)}{\lambda_2 - \lambda_1}$$

20. **Ans. (C)**

Sol. $\frac{d}{2} = (\mu - 1)t \frac{D}{d}$
 $(\mu - 1)t = \frac{d^2}{2D}$

$$\Delta x = (\mu - 1)t = \frac{d^2}{2D}$$

SECTION-II

1. **Ans. 2.00**

Sol. $\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2} = \sqrt{\frac{g}{l} - \left(\frac{6\pi\eta r}{2m}\right)^2}$

2. **Ans. 30.00**

Sol. $B = \mu_0 H = \mu_0 ni$

$$n \times 10^2 = \frac{100}{10 \times 10^{-2}} \times 3$$

$$n = 30$$

3. **Ans. 45.00**

Sol. $\tan \theta = \tan \theta' \cos \alpha$

$$(\because \theta' = \tan^{-1}\left(\frac{2}{\sqrt{3}}\right), \alpha = 30^\circ)$$

$$\therefore \tan \theta = \tan\left(\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)\right) \cos 30^\circ$$

$$\tan \theta = \frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{2} = 1 \Rightarrow \theta = 45^\circ$$

4. **Ans. 37.50**

Sol. Initially the polaroids are crossed to each other, that is the angle between their polarising directions is 90° . When both are rotated through 30° opposite sense, then the angle between their polarising directions will become 30° .

Let the intensity of the incident unpolarised light = I_0

Then the intensity of light emerging from

$$\text{the first polaroid is } I_1 = \frac{1}{2} I_0$$

This light is plane polarised and passes through the second polaroid.

The intensity of light emerging from the second polaroid is $I_2 = I_1 \cos^2 \theta$

θ = the angle between the polarising directions of the two polaroids.

$$I_1 = \frac{1}{2} I_0 \quad \text{and } \theta = 30^\circ$$

$$\text{so } I_2 = I_1 \cos^2 30^\circ = \frac{1}{2} I_0 \cos^2 30^\circ$$

$$\Rightarrow \frac{I_2}{I_0} = \frac{3}{8}$$

$\therefore$ transmission percentage

$$= \frac{I_2}{I_0} \times 100 = \frac{3}{8} \times 100 = 37.5\%$$

5. **Ans. 25.00**

Sol. $R.P = \frac{d}{1.22\lambda}$

where $d \rightarrow$ diameter

$$\therefore n = 25$$

6. **Ans. 5.00**

7. **Ans. 21.70 to 21.75**

Sol. Percentage modulation

$$= \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \times 100 = 21.739 \approx 21.74$$

8. **Ans. 2.20 to 2.25**

Sol. $I = \frac{1}{2} \epsilon_0 E_0^2 c = \frac{P}{4\pi r^2}$

$$E_0 = \sqrt{\frac{P}{2\pi r^2 \epsilon_0 c}} = \sqrt{\frac{p}{2r^2}} \frac{1}{\sqrt{\pi \epsilon_0 c}}$$

$$x = \sqrt{\frac{P}{2r^2}} = \sqrt{5} = 2.24$$

9. Ans. 6.41

Sol. Least count

$$= \frac{\text{pitch}}{\text{No. of divisions}} = \frac{2}{200} = 0.01 \text{ mm}$$

$$\text{Reading} = 3 \times 2 + (46-5) (0.01) = 6.41 \text{ mm}$$

10. Ans. 4.00

Sol. $f = \frac{v}{2(\ell + 2e)}$ where e = end correction
 $= 0.6 \text{ r}$

$$\therefore f = \frac{v}{2(\ell + 2 \times 0.6r)} = \frac{v}{2(\ell + 1.2r)}$$

$$\therefore \frac{\Delta f}{f} = \frac{\Delta v}{v} - \frac{\Delta(\ell + 1.2r)}{\ell + 1.2r}$$

$$= \frac{\Delta v}{v} - \frac{\Delta \ell + 1.2 \Delta r}{\ell + 1.2r}$$

here $\frac{\Delta v}{v} = 0$ (given) $\frac{\Delta f}{f} \times 100$

$$= - \frac{\Delta \ell + 1.2 \Delta r}{\ell + 1.2r} \times 100$$

for maximum % error : $\Delta \ell = 0.1$, $\Delta r = 0.05$

$$\left(\frac{\Delta f}{f} \times 100 \right)_{\text{max}} = \frac{0.1 + 1.2 \times 0.05}{94 + 1.2 \times 5} \times 100$$

$$= 0.16\%$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

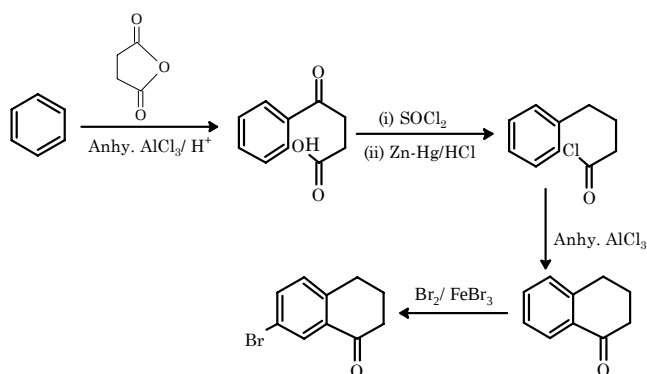
1. Ans. (B)

2. Ans. (A)

3. Ans. (C)

4. Ans. (B)

5. Ans. (D)



6. Ans. (A)

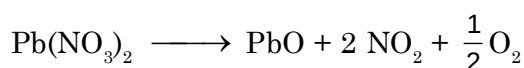
$$\Delta S_{\text{Reaction}} = [186.264] - [5.740 + 2 \times 130.684]$$

or $\Delta S_{\text{System}} = -80.844 \text{ J/K}$

$$\Delta S_{\text{Surrounding}} = + \frac{\Delta H}{T} = \frac{74.81 \times 10^3}{298} = 251 \text{ J/K}$$

$$\Delta S_{\text{Universe}} = -80.844 + 251 = 170.1 \text{ J/K}$$

7. Ans. (D)



8. Ans. (A)

Vicinal diol compounds breakdown by HIO_4 .

9. Ans. (C)

(A) θ incorrect less than $109^\circ 28'$

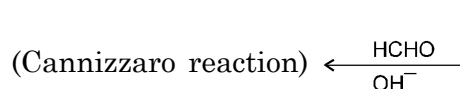
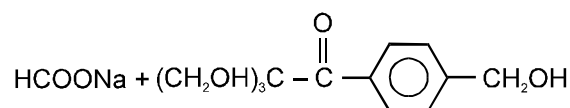
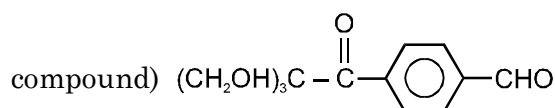
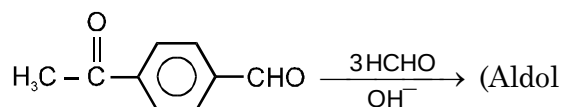
(B) incorrect position of lone pair

(C) correct square planar structure with $\theta = 90^\circ$

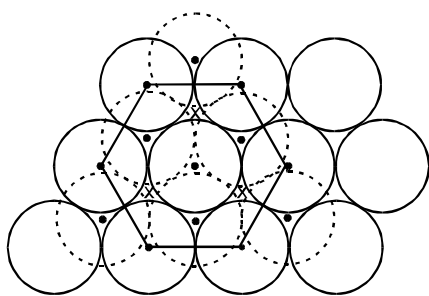
(D) θ incorrect because it is square planar with two lone pair at Xe and $\theta = 90^\circ$

10. Ans. (B)

3 moles of HCHO give cross aldol condensation with aromatic ketone one mole of HCHO under goes cross Cannizzaro reaction with the aldol product.



11. Ans. (B)



hcp = AB AB AB pattern repeat

For calculating voids between two layers A and B.

The dots are representing tetrahedral voids.

$$\text{no. of tetrahedral voids} = 3 + 1 + \frac{1}{3} \times 6 = 6$$

so in one complete unit cell total tetrahedral voids = 12, out of which 8 are completely inside but rest are shared by other unit cells.

Octahedral voids = 3 = represented by (x).

So total octahedral voids = 6 = All are completely inside. $[4R = \sqrt{3}a]$

12. Ans. (A)

13. Ans. (B)

14. Ans. (A)

Adsorption of gas on solid surface.

15. Ans. (B)

16. Ans. (B)

The given two compounds are stereoisomers and they are not mirror image of each other hence they are diastereomers.

17. Ans. (D)

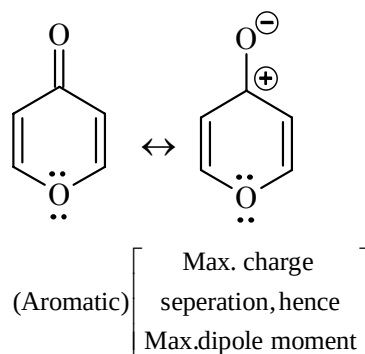
$$\lambda = \frac{h}{\sqrt{2m_p eV}} \quad \lambda_\alpha = \frac{h}{\sqrt{2m_\alpha (2e)V_\alpha}}$$

$$= 2m_p eV = 2.4 \times m_p \times 2eV_\alpha$$

$$\Rightarrow V_\alpha = \frac{V}{8}$$

18. Ans. (B)

19. Ans. (A)



20. Ans. (A)

$$32 \times 5 \times 5 = 250 \times M \times 1 \Rightarrow M = 3.2$$

SECTION-II

1. Ans. (3.00)

2. Ans. (4.00)

The given compound

has two stereocentres therefore it has four stereoisomers.

3. Ans. (140.00)

Charge passed = 0.01 Faraday

At the anode $\left(\text{H}_2\text{O} \longrightarrow \frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \right)$

with 90 % efficiency 0.01 $\times$ 0.9 F have been

used and will produce $\frac{1}{4} \times 0.01 \times 0.9$ mole of O_2 i.e. 0.00225 mol O_2 .

At the cathode $2\text{H}_2\text{O} \xrightarrow{+2\text{e}^-} \text{H}_2 + 2\text{OH}^-$

$$\text{moles of H}_2 \text{ produced} = \frac{0.01 \times 0.8}{2} \text{ mol}$$

$$= 0.004 \text{ mol}$$

Total moles produced of gases

$$= 0.004 + 0.00225 = 0.00625 \text{ mol}$$

$$\text{vol. at STP} = 0.00625 \times 22400 \text{ mL} = 140 \text{ mL}$$

4. Ans. (281.00)

Net stabilization energy of Fe^{2+} with CN^-

$$\text{ligands } t_{2g}^{2,2,2} e_g^{0,0} = -2.4 \Delta_0 + 2P$$

$$= (-2.4 \times 25000 + 2 \times 15000) \text{ cm}^{-1}$$

$$= -30000 \text{ cm}^{-1}$$

net stabilisation energy of Fe^{2+} ions with

$$\text{H}_2\text{O ligands } t_{2g}^{2.11} e_g^{1.1} = -0.4 \Delta_0$$

$$= 4000 \text{ cm}^{-1}$$

so required difference = 26000 cm^{-1}

$$= 26 \times 10^5 \text{ m}^{-1}$$

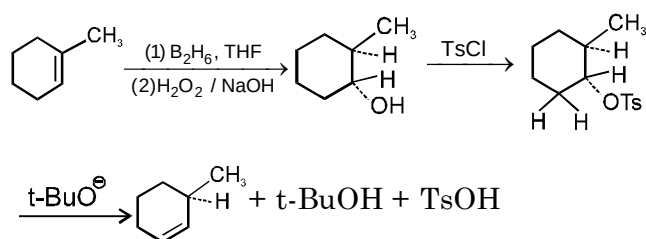
$$= 6 \times 10^{-34} \times 3 \times 10^8 \times 26 \times 10^5 \times 6 \times 10^{23} \times 10^{-3} \text{ kJ/mole}$$

$$= 2808 \times 10^{-1} \text{ kJ/mole}$$

$$= 281 \text{ kJ/mole}$$

$$= 280.8 \text{ kJ/mol}$$

5. Ans. (3.00)



6. Ans. (2.50)

$$PV = nRT$$

$$n_{\text{N}_2} = \frac{10}{R \times T} \times \frac{80}{1000}$$

$$P = K_H x = K_H \times \frac{n_{\text{N}_2}}{n_{\text{H}_2\text{O}}}$$

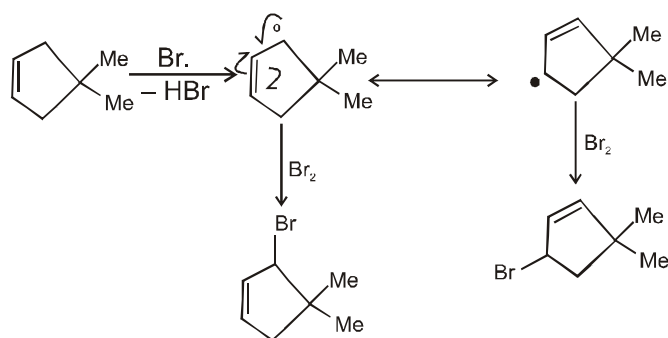
$$\frac{10}{20} = \frac{K_H \times \frac{10}{RT} \times \frac{80}{1000}}{K_H \times \frac{20}{RT} \times \frac{40}{1000}} \times \frac{V}{\frac{18}{(5000)}}$$

$$\frac{1}{1} = 2 \times \frac{V}{5000} \Rightarrow V = 2.5 \text{ litre}$$

7. Ans. (5.00)

Product is SO_2

8. Ans. (4.00)



9. Ans. (125.00)

$$\Delta H = \Delta H_1 \times \frac{3}{4} + \Delta H_2 \times \frac{1}{4}$$

$$= 100 \times \frac{3}{4} + 200 \times \frac{1}{4} = 125 \text{ cal}$$

10. Ans. (6.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A)

$$\hat{a} + \hat{b} = \left(\frac{7\hat{i} - 4\hat{j} - 4\hat{k}}{9} \right) + \left(\frac{2\hat{i} - \hat{j} + 2\hat{k}}{3} \right)$$

$$= \frac{(7\hat{i} - 4\hat{j} - 4\hat{k}) + (-6\hat{i} - 3\hat{j} - 6\hat{k})}{9} = \frac{\hat{i} - 7\hat{j} + 2\hat{k}}{9}$$

$$\therefore \text{Vector mag. } 3\sqrt{6} = \frac{3\sqrt{6}(\hat{i} - 7\hat{j} + 2\hat{k})}{\sqrt{54}}$$

2. Ans. (C)

$$f(g(x)) = 1 + 3 \left(\frac{x - 2x^2 + 5x^3}{3} \right)$$

$$\Rightarrow g(x) = \frac{5x^3 - 2x^2 + x}{3}$$

$$\therefore 3g(1) = 4$$

3. Ans. (A)

$$f(x) \text{ is continuous when } x^3 + 3x = 3x^2 + 1$$

$$\Rightarrow x = 1$$

Hence (A)

4. Ans. (D)

$$\int_0^{\pi/2} x(1)dx + \int_{\pi/2}^{3\pi/2} x(-1)dx + \int_{3\pi/2}^{2\pi} x(1)dx$$

$$\left(\frac{x^2}{2} \right)_0^{\pi/2} - \left(\frac{x^2}{2} \right)_{\pi/2}^{3\pi/2} + \left(\frac{x^2}{2} \right)_{3\pi/2}^{2\pi}$$

$$\frac{\pi^2}{4} - \frac{9\pi^2}{4} + \frac{\pi^2}{4} + 4\pi^2 - \frac{9\pi^2}{4} = 0$$

5. Ans. (B)

AAAE, IPL

ways = total - IPL together

$$N = ({}^8C_5 \cdot 1)3! - ({}^6C_1 \cdot 1)$$

$$N = 330$$

6. Ans. (B)

$$f(x) = \begin{cases} (x-1)^3 + 1; & -1 < x \leq 1 \\ 5x - 4; & 1 < x \leq 2 \end{cases}$$

$$g(x) = x^2 + 1$$

$$f(g(x)) = (g(x) - 1)^3 + 1, \quad -1 < g(x) \leq 1$$

$$5g(x) - 4, \quad 1 < g(x) \leq 2$$

$$-1 < g(x) \leq 1 \Rightarrow x = 0$$

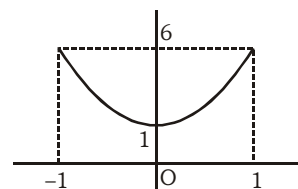
$$1 < g(x) \leq 2 \Rightarrow 1 < x^2 + 1 \leq 2$$

$$\Rightarrow 0 < x^2 \leq 1$$

$$\Rightarrow -1 \leq x \leq 1, x \neq 0$$

$$f(g(x)) = \begin{cases} (x^2)^3 + 1; & x = 0 \\ 5(x^2 + 1) - 4; & -1 \leq x \leq 1, x \neq 0 \end{cases}$$

$$f(g(x)) = \begin{cases} 1, & x = 0 \\ 5x^2 + 1, & -1 \leq x \leq 1, x \neq 0 \end{cases}$$



so range [1, 6]

7. Ans. (D)

Let 60 be divided in two parts x and $60 - x$ such that $f(x) = (60 - x)x^2$ is maximum

$$f(x) = 60x^2 - x^3$$

for maximum value of $f(x)$

$$f'(x) = 0 \text{ and } f''(x) < 0$$

$$120x - 3x^2 = 0$$

$$x = 0 \text{ or } x = 40 \quad f''(40) < 0$$

$\Rightarrow f(x)$ is maximum when $x = 40$

$$x = 40, \quad 60 - x = 20$$

$$\text{Required ratio } \frac{60-x}{x} = \frac{20}{40} = 1:2$$

8. Ans. (A)

Any plane through the line $x + y + z + 3 = 0$ $3x + y - z$ will be

$$3x + y - z + \lambda(x + y + z + 3) = 0$$

$$(3 + \lambda)x + (1 + \lambda)y + (\lambda - 1)z + 3\lambda = 0 \dots (1)$$

this plane is parallel to the line

$$\frac{x-2}{1} = \frac{y+3}{2} = \frac{z-5}{3}$$

$$\Rightarrow (3 + \lambda)1 + (1 + \lambda)2 + (\lambda - 1)3 = 0$$

$$6\lambda + 2 = 0; \quad \lambda = -\frac{1}{3}$$

Put value of λ in

$$8x + 2y - 4z - 1 = 0$$

$\Rightarrow$ Direction ratios of normal to this plane

$$8 : 2 : -4 = 4 : 1 : -2$$

9. Ans. (C)

p = India wins the toss

q = Rohit will make century

Given statement is $p \rightarrow q$

and it is logically equivalent to

$\sim q \rightarrow \sim p$ (contrapositive)

$\Rightarrow$ correct ans is

If Rohit does not make century, then India will not win.

10. Ans. (B)

$$\frac{dy}{dx} + \left(-1 - \frac{f'(x)}{f(x)} \right) y = f(x)$$

$$\text{I.F.} = e^{-\int \left(1 + \frac{f'(x)}{f(x)} \right) dx} = e^{-x - \ln f(x)} = \frac{e^{-x}}{f(x)}$$

$$\Rightarrow y \cdot \frac{e^{-x}}{f(x)} = \int e^{-x} dx$$

$$\Rightarrow y = -f(x) + ce^x f(x)$$

11. Ans. (A)

A line will be tangent to a circle, if length of perpendicular drawn from centre $(a, 0)$ upon the line $lx + my + n = 0$ is equal to radius $= a$

$$\Rightarrow \left| \frac{al + n}{\sqrt{l^2 + m^2}} \right| = a$$

$$\Rightarrow (al + n)^2 = a^2 (l^2 + m^2)$$

$$\Rightarrow n^2 + 2aln = a^2 m^2$$

$$\Rightarrow n^2 + 2aln - a^2 m^2 = 0$$

12. Ans. (A)

$$A = \int_0^\pi \left(\frac{x}{1 + \sin x} \right)^2 dx; \quad I = \int_0^\pi \frac{x^2 (1 + \cos x)}{(1 + \sin x)^2};$$

$$I = A + \int_0^\pi \frac{x^2 (\cos x)}{(1 + \sin x)^2};$$

$$A + \underbrace{\left(\frac{-x^2}{1+\sin x} \right)_0^\pi + 2 \int_0^\pi \frac{x}{1+\sin x} dx}_{I'} ;$$

$$I' = \int_0^\pi \frac{x}{1+\sin x} dx$$

$$\text{using king } I' = \int_0^\pi \frac{\pi-x}{1+\sin x} dx ;$$

$$2I' = 2\pi \int_0^{\pi/2} \frac{dx}{1+\sin x}$$

$$I' = \pi \int_0^{\pi/2} \frac{1-\sin x}{\cos^2 x} dx ; I'$$

$$= \pi (\tan x - \sec x)_0^{\pi/2} ; I'$$

$$= \pi [\lim_{x \rightarrow \pi/2} (\tan x - \sec x) - (0-1)] = \pi$$

$$\therefore I = A + 2\pi - \pi^2]$$

13. Ans. (D)

$$\text{Put } 1-x^{2013} = t^2$$

$$\Rightarrow -2013x^{2012}dx = 2tdt$$

$$\Rightarrow dx = \frac{-2}{2013} \frac{tdt}{x^{2012}}$$

$$\therefore = \int \frac{-2}{2013} \frac{tdt}{(1-t^2)t}$$

$$= \frac{-2}{2013} \int \frac{dt}{1-t^2}$$

$$= \frac{1}{2013} \log \left| \frac{t-1}{t+1} \right| + C$$

$$= \frac{1}{2013} \log \left| \frac{\sqrt{1-x^{2013}}-1}{\sqrt{1-x^{2013}}+1} \right| + C$$

14. Ans. (C)

Let a, b, c are a, ar, ar²

$$\Rightarrow a = \frac{3}{\frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2}} \Rightarrow a + \frac{1}{r} + \frac{1}{r^2} = 3$$

$$\Rightarrow 3r^2 + r^2 + r + 1$$

$$\Rightarrow 2r^2 - r - 1 = 0$$

$$\begin{cases} r=1 & (\text{rejected } \because a, b, c \text{ are diff.}) \\ a = -\frac{1}{2} \end{cases}$$

$$\Rightarrow a, b, c \text{ are } a, -\frac{a}{2}, \frac{a}{4}$$

15. Ans. (B)

$$\frac{dy}{dx} + x \tan y = x \tan y \cdot \sin y$$

$$\Rightarrow \cos y \frac{dy}{dx} + x \sin y = x \sin^2 y$$

$$\text{Let } \sin y = t$$

$$\Rightarrow \frac{dt}{dx} + tx = t^2 x$$

$$\Rightarrow \frac{1}{t^2} \frac{dt}{dx} + \frac{x}{t} = x$$

$$\text{Let } \frac{1}{t} = u$$

$$\Rightarrow u \cdot e^{-x^2/2} = - \int x \cdot e^{-x^2/2} dx = e^{-x^2/2} + c$$

$$\Rightarrow u = 1 + ce^{x^2/2}$$

$$\Rightarrow \sin y (1 + ce^{x^2/2}) = 1$$

$$\Rightarrow k = 1$$

16. Ans. (A)

$$OC = OA \cos \frac{\pi}{3} = 5$$

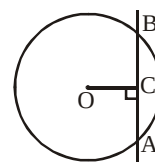
$$y - 7 = m(x - 1)$$

$$OC = \frac{|m-7|}{\sqrt{1+m^2}} = 5$$

$$\Rightarrow m = \frac{3}{4}, -\frac{4}{3}$$

$$\therefore \text{Equation of AB is } 4y - 3x = 25$$

$$\text{or } 3y + 4x = 25$$



17. Ans. (B)

Since line joining AP is not parallel to the plane

So, it intersects the plane.

$$\frac{x-1}{0} = \frac{y-0}{5} = \frac{z+3}{-10} = r$$

$$x = 1, y = 5r, z = -10r - 3$$

which lies on the plane.

$$2x + 3y + 5z = 1$$

$$\Rightarrow 2 + 15r - 50r - 15 = 1$$

$$\Rightarrow r = \frac{-14}{35} = -\frac{2}{5}$$

$$\therefore P \equiv (1, -2, 1)$$

$$AP = \sqrt{0+4+16} = 2\sqrt{5}$$

18. Ans. (C)

The curve cuts x-axis at (1, 0)

$$y' = (\sqrt{5-x^6}) \cdot 3x^2 - (\sqrt{5-x^4}) 2x$$

6. **Ans. 7.00**

$$N = 2^{1999} \cdot 5^{2001} = 2^{1999} \cdot 5^{1999} \cdot 5^2 = (10)^{1999} \cdot 25$$

Sum of digits is 7

7. **Ans. 13.33 or 13.34**

Ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$

End of LR = $(3, \frac{16}{5})$

tangent at $(3, \frac{16}{5})$ will be $T = 0$

$$\frac{3x}{25} + \frac{y}{5} = 1$$

Sum of intercepts = $\frac{25}{3} + 5$

$$= 8.33 + 5 = 13.33$$

8. **Ans. 2.50**

$$f(x) = 2x^2 - \ln x$$

For increasing $f'(x) > 0$

$$4x - \frac{1}{x} > 0 \Rightarrow \frac{4x^2 - 1}{x} > 0$$

$$\Rightarrow x \in \left(\frac{1}{2}, \infty\right)$$

$$a = \frac{1}{2} \therefore a + \frac{1}{2} = \frac{1}{2} + 2 = 2.50$$

9. **Ans. 58.00**

Let the weight of the teacher is w kg, then

$$40 + \left[\frac{1}{2}\right] = \frac{[35 + 40 + w]}{[35 + 1]}$$

$$36 \times 40 + 36 \times 12 = 35 \times 40 + w$$

$$w = 58$$

Weight of the teacher = 58 kg.

10. **Ans. 1.00**

$$|A| = -1$$

$$\frac{1}{-2011} |A^5 - 201A^4| = \frac{|A|^4}{-2011} |A - 201.I|$$

$$= \frac{1}{-2011} \begin{vmatrix} 0 & 2011 \\ 1 & -191 \end{vmatrix} = \frac{-2011}{-2011} = 1$$